

INPhO 2003

1. (a) **An elastic collision** [4]

A mass m_1 moving horizontally on a frictionless floor with velocity v_0 strikes head-on a massless spring (of spring constant k) connected to a mass m_2 . Determine the maximum compression of the spring.
- (b) **Disc slipping before rolling** [12]

A uniform disc of mass m and radius R is spinning with angular speed ω_0 about its axis, as it is placed on a horizontal floor. The coefficient of static friction between the disc and the floor is μ_s and that of kinetic friction is μ_k . The disc slips for some time and then rolls without slipping with its centre of mass speed v_f . It will be interesting to obtain v_f by three different ways:

 - (i) Determine the time for which the disc slips until it begins to roll. Hence determine v_f .
 - (ii) Determine the work done by the frictional force for the entire motion. Then use the work-energy theorem to determine v_f .
 - (iii) Use the principle of conservation of angular momentum to determine v_f .
- (c) **Period of revolution of a binary star system** [4]

Each star in a binary star system has the same mass as that of the Sun, and the distance between the two stars equals the Sun-Earth distance. What is the period of revolution (in terms of the Earth year)?
2. (a) **Efficiency of a heat engine** [10]
 - (i) A heat engine using an ideal gas as a working substance uses the following cycle:
 - (I) The gas is expanded adiabatically from (p_1, V_1) to (p_2, V_2) .
 - (II) It is compressed isobarically to (p_2, V_1) .
 - (III) Its pressure is increased to p_1 at constant volume.Determine the efficiency of the engine in terms of p_1, V_1, p_2, V_2 and $\gamma (= c_p/c_v)$ and show that it is less than 1, as expected.
 - (ii) What is the value of γ if the working substance is a nitrogen gas? Assume that a nitrogen molecule has no vibrational degree of freedom.
- (b) **Lifting a container** [10]

A small spherical container of radius R and thickness d has hydrogen gas stored at high pressure p ($p \gg p_0$, where p_0 is the atmospheric pressure). A light balloon is fixed on the top of the container and the gas is slowly introduced into the balloon causing it to expand until the pressure of the gas equals p_0 . The density of the wall material is ρ and that of air is ρ_{air} . The tensile strength, that is, the restoring force per unit cross sectional area of the wall is σ . Determine the minimum value of σ/ρ (in terms of other parameters in the problem) required for the balloon to lift the container. Neglect the weight of the balloon and the gas inside it and assume that the temperature of the gas remains constant.
3. (a) **The moon's albedo** [10]

The full moon is a delightful sight, but we know that its brightness is due to reflection of sunlight. The reflection coefficient (fraction of incident power that is reflected) of the moon is called its albedo which we estimate in this problem.

 - (i) The intensity of sunlight on the Earth at noon is about 1400 Wm^{-2} . Estimate the power output of the Sun.

- (ii) The brightness of astronomical objects is measured in terms of a dimensionless quantity called the magnitude m , defined as

$$m = -2.5 \log_{10} \left(\frac{I}{I_0} \right)$$

where I is the intensity of light received from the object at the Earth and I_0 is a reference intensity: $I_0 = 2.52 \times 10^{-8} \text{ Wm}^{-2}$. (The dimmest star visible to the naked eye has a magnitude of about 5.) What is the magnitude of the Sun?

- (iii) The magnitude of the full moon is -12.5 . What is the moon's albedo?
You are given that

$$\begin{aligned} \text{Average Earth-Sun distance} &= 1.5 \times 10^{11} \text{ m} \\ \text{Average Earth-moon distance} &= 3.8 \times 10^8 \text{ m} \\ \text{Radius of the moon} &= 1.74 \times 10^6 \text{ m} \end{aligned}$$

Neglect absorption of light as it travels through space.

(b) **Telescopic resolution**

A space telescope orbiting the Earth has a mirror diameter of 3 m. Estimate the distance between two stars which are 10^5 light years away from the Earth and can just be resolved by the telescope.

(c) **Slide Projector**

A slide with a $24 \text{ mm} \times 36 \text{ mm}$ picture is projected on a screen placed 12 m from the slide. The image of the slide picture on the screen measures $1.0 \text{ m} \times 1.5 \text{ m}$. Determine the location of the projection lens and its focal length.

4. (a) **Conversion factor between SI and Gaussian CGS units.**

The SI units are familiar to you. It is often useful to know the conversion factors between different systems of units. Derive the conversion factor between the units of potential in SI and Gaussian CGS system:

$$1 \text{ V} = (?) \text{ statvolt}$$

Remember that in Gaussian CGS units, the proportionality constant in Coulomb's law is taken to be 1 (and dimensionless), i.e. it reads:

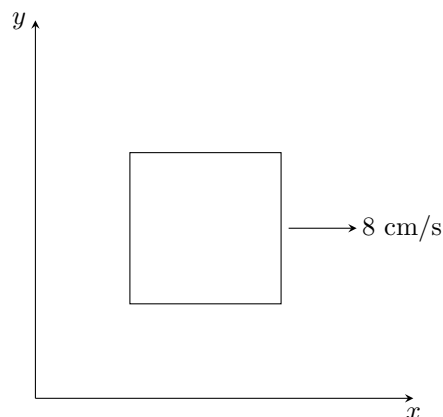
$$F = \frac{q_1 q_2}{r^2} \quad (\text{Gaussian CGS})$$

(CGS stands for cm gm sec.)

The definition of potential in terms of work per unit charge is the same in the two systems.

(b) **Induced current in a loop**

A square loop of side 12 cm with its sides parallel to x - and y -axes is moved with a velocity of 8 cm/s in the positive x -direction in an environment containing magnetic field in the positive z -direction. The field is neither uniform in space nor constant in time. It increases by 10^{-3} T per cm as one moves in the negative x -direction, and it is decreasing in time at the rate of 10^{-3} T s^{-1} . Determine the direction (with reference to the figure given) and magnitude of the induced current in the loop if its resistance is $4.5 \text{ m}\Omega$.



(c) **Charge separation in a rotating cylinder**

[10]

A solid cylindrical conductor is placed in a uniform magnetic field $\vec{B}$ parallel to its axis of symmetry. The conductor is rotated with angular velocity $\vec{\omega}$ about its axis of symmetry. ($\vec{\omega} \parallel \vec{B}$). Determine the charge density inside the rotating cylinder. For what angular velocity is there no separation between the positive charges and negative charges (electrons)? (Denote the charge of electron by $-e$ ($e > 0$) and its mass by m).

5. (a) **Pulse on a string**

[10]

A horizontal string of linear mass density 0.5 g cm^{-1} and length 30 cm is tied to a rigid wall at one end and to a frictionless ring at the other end. The ring can move on a vertical rod. A pulse on the string moves towards the ring at the speed of 20 cm s^{-1} . The pulse is symmetric about its maximum which is initially located at 20 cm from the ring end. Assume that the wave is reflected from the ends without any loss of energy.

- (i) Determine the tension the string.
- (ii) Find the time taken by the string to regain its initial shape.
- (iii) Find the time taken by the pulse to arrive at the initial location, with the same initial shape and direction of velocity.

(b) **Radioactive carbon dating**

[10]

- (i) A vessel of volume 250 cm^3 contains carbon dioxide gas at a pressure of $1.5 \times 10^4 \text{ Pa}$ and temperature 300 K . How many β -decay counts would be recorded in a week? It is known that radioactive carbon (^{14}C) has a half life of 5730 y and has an abundance of 1 part in 10^{12} of normal carbon in the atmosphere.
- (ii) An old sample of wood is burnt and the carbon dioxide produced is filled in the same vessel at the same temperature and pressure. Over one week, 1100 counts are recorded. Estimate the age of the sample.